

Parallelogram Cut-and-Slide Lab

Unit 5: Architecture Firm • Lesson 5-1 • EduWonderLab

UNIT 5 • LESSON 5-1 • TEACHER NOTES

Standard 6.G.1

FORMAT Hands-on cut-and-rearrange lab

TIME 25–30 min

GROUPING Pairs

TOPIC Area of Parallelograms

STANDARD 6.G.1

PREP LEVEL Medium — print parallelogram templates; scissors and glue

Learning goal *I can find the area of a parallelogram using base and height.*

Teacher setup

- Print one template page per pair. Students cut a parallelogram and slide the triangle to form a rectangle.
- They measure base and height, then solve the task cards.
- Stress that height is perpendicular to the base, not the slanted side.

Materials

Parallelogram templates, Scissors, Glue, Area recording sheet, Pencil.

Differentiation & language support

- Show the right-angle mark for height on each figure (SPED).
- Sentence stem: “The height is ____, not ____, because ____.”
- ESOL: label base and height on a model figure first.

Key vocabulary (English / Español):

Term	Español	What it means
base	base	a side of the figure you measure from
height	altura	the perpendicular distance to the base
perpendicular	perpendicular	meeting at a right angle

Quick teacher check: *Watch for students who use the slanted side instead of height (Card 8).*

Parallelogram Cut-and-Slide Lab

Unit 5: Architecture Firm · Lesson 5-1 · EduWonderLab

Name: _____ Date: _____ Class: _____

Your goal: *I can find the area of a parallelogram using base and height.*

What you need

Parallelogram templates, Scissors, Glue, Recording sheet.

How to play

1. Cut your parallelogram and slide the triangle to make a rectangle.
2. Identify the base and the height (perpendicular distance).
3. Use $\text{Area} = \text{base} \times \text{height}$ on each card.
4. Record the area with square units.

Area Cards

Cut and slide to turn a parallelogram into a rectangle, then use $\text{Area} = \text{base} \times \text{height}$.

1. base = 6 cm, height = 4 cm	2. base = 8 cm, height = 3 cm
3. base = 10 cm, height = 5 cm	4. base = 7 cm, height = 6 cm
5. base = 12 cm, height = 2.5 cm	6. base = 9 cm, height = 4 cm
7. base = 5.5 cm, height = 4 cm	8. Trap card: base = 8 cm, slanted side = 5 cm, height = 4 cm. Area?

Record your work (base $\times$ height):

Explain your thinking

Explain why a parallelogram and a rectangle with the same base and height have the same area.

Sentence frame: *The height is ___, not ___, because height must be ___ to the base.*

★ **Challenge:** Draw two different parallelograms that both have an area of 24 cm^2 .

ANSWER KEY

Teacher copy — please do not distribute to students.

UNIT 5 • LESSON 5-1 • Parallelogram Cut-and-Slide Lab

Standard 6.G.1

Answers

1. base = 6 cm, height = 4 cm Answer: 24 cm ²	2. base = 8 cm, height = 3 cm Answer: 24 cm ²
3. base = 10 cm, height = 5 cm Answer: 50 cm ²	4. base = 7 cm, height = 6 cm Answer: 42 cm ²
5. base = 12 cm, height = 2.5 cm Answer: 30 cm ²	6. base = 9 cm, height = 4 cm Answer: 36 cm ²
7. base = 5.5 cm, height = 4 cm Answer: 22 cm ²	8. Trap card: base = 8 cm, slanted side = 5 cm, height = 4 cm. Area? Answer: 32 cm ² (use height 4, not 5)

Teaching notes & look-fors

- Card 8 tests the common error: use the perpendicular height (4), not the slanted side (5).
- Area = base × height; cutting and sliding shows why.